

Switching Real Power

A logic pin can't switch a real load — a MOSFET can. Drive a relay or solenoid safely, with the flyback diode that keeps the kick from killing your board.

ESP32-005 · ~45 MIN · INTERMEDIATE

1. The one idea

A 3.3V logic pin can switch a few milliamps — not a motor, a solenoid, or a relay coil. A **logic-level MOSFET** lets a tiny gate voltage switch real current, and a **flyback diode** absorbs the inductive kick that would otherwise reset (or destroy) your board. This is how a controller touches power safely.

2. Why it matters

Every actuator past an LED — relays, solenoids, motors, pumps, valves — pulls more current than a GPIO can give and, if it has a coil, fights back when you switch it off. The collapsing magnetic field dumps a high-voltage spike backward into your circuit. Get the MOSFET and the flyback diode right and you can drive anything; get them wrong and you'll spend an afternoon wondering why the ESP32 keeps rebooting.

3. Parts & wiring

- ESP32 dev board
- IRLZ44N logic-level MOSFET (low-side switch)
- A coil load — 2-channel relay module **or** the 12V solenoid
- 1N4007 flyback diode
- 100–220Ω gate resistor, 10kΩ gate pulldown
- Bench PSU at the load's voltage (5V for the relay, 12V for the solenoid)

Node	Connect	Why
ESP32 GPIO25	→100Ω→ MOSFET gate	series resistor limits gate inrush
MOSFET gate	→10kΩ→ GND	pulldown: the load fails OFF at boot/reset
MOSFET source	GND (shared with PSU)	low-side switch
MOSFET drain	load – terminal	the MOSFET sinks the load current

Node	Connect	Why
Load + terminal	PSU +	load runs from its own supply
1N4007	across the load, cathode to PSU +	clamps the turn-off spike

*Three rules that aren't optional. **Flyback diode** across every coil, band (cathode) to the positive rail — without it, turning the coil off spikes hundreds of volts backward. **Gate pulldown** so the load is OFF while the ESP32 boots or resets, not chattering. **Separate supply** for the load; share only ground. Note: the IRLZ44N is logic-level but 3.3V is the low end of its gate drive — fine for these small coils, but it will warm up; for heavy loads add a gate driver or a higher gate voltage.*

4. Predict (do this before uploading)

- At the moment you switch the coil OFF, which way does the voltage spike, and what stops it?
- During the ESP32's boot, before your code runs, is the load ON or OFF? What decides that?
- Without the flyback diode, what's the first symptom you'll see?

5. Build it — switch, and switch off cleanly

```
const int GATE = 25;           // → 100 ohm → MOSFET gate (10k pulldown to GND)

void setup() {
  pinMode(GATE, OUTPUT);
  digitalWrite(GATE, LOW);    // start OFF – match the pulldown
}

void loop() {
  digitalWrite(GATE, HIGH);   // load ON
  delay(1000);
  digitalWrite(GATE, LOW);    // load OFF – the flyback diode catches the kick here
  delay(1000);
}
```

The code is trivial; the *circuit* is the lesson. The interesting instant is the falling edge — `digitalWrite(GATE, LOW)` — because that's when the coil's field collapses and the diode earns its place. Listen to the relay click cleanly, with no reboot.

6. Test against your prediction

Power the load from the bench PSU, watch its current on the meter, and toggle. The relay clicks (or the solenoid pulls) once per second, and the ESP32 stays alive across every switch-off. Now — only if you're curious and the spike is small — probe the drain on the Analog Discovery 3 at turn-off: with the diode, a clean clamp; the spike you *don't* see is the one that isn't hitting your board.

7. What goes wrong (pre-loaded debugging)

- **ESP32 reboots every time the load turns off** → no (or backward) flyback diode. Band to the positive rail.
- **Load is on at power-up before your code runs** → missing gate pulldown; the floating gate drifted high. Add the 10k Ω .
- **MOSFET gets hot** → 3.3V only partly enhances it for a biggish load; reduce current, add a gate driver, or drive the gate at 5V.
- **Load won't switch at all** → you wired it high-side (load on the source side); the IRLZ44N is an N-channel low-side switch — load between +V and drain.
- **Everything browns out** → load sharing the ESP32's supply. Give it the bench PSU; share only ground.

8. Checkpoint

You switched a real, coil-driven load from a logic pin — safely. The MOSFET carries the current the pin can't, the gate network makes it fail OFF, and the flyback diode tames the inductive kick. That trio is the gateway to every powered actuator: pumps, valves, big LEDs, motors. Respect the falling edge and you can drive almost anything.

9. Push further

- Swap the on/off for PWM (`analogWrite` on the gate) to dim a 12V LED strip or ramp a motor.
- Replace the 1N4007 with a faster Schottky (1N5819) for PWM'd loads — cooler, quieter switching.
- Drive the gate from a dedicated gate-driver IC and watch the turn-on edge sharpen on the scope.